ONGC

Vanna AI Platform

AI-Powered Database Intelligence for ONGC

Complete User Guide & Platform Walkthrough

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Confidential — Internal Use Only

Platform URL	`http://localhost`
Admin Email	`admin@vanna-platform.local`
Admin Password	`Admin@ONGC123`
Role	`super_admin (full access)`

Platform Overview

Vanna AI is a centralized AI-powered database query platform built for ONGC. It lets any authorised user query connected enterprise databases using plain English — no SQL knowledge required. The AI converts the question into optimised SQL, executes it, and returns results as a table you can export to CSV.

Key Features

- **Natural Language to SQL** — Type any question in English; Vanna AI writes the SQL automatically
- **Multi-database support** — PostgreSQL, Oracle, MySQL / MariaDB, SQL Server, MongoDB (Phase-1 experimental)
- **Role-Based Access Control (RBAC)** — 4 roles controlling exactly what each user can see and do
- **Full audit trail** — Every query logged with user, time, generated SQL, and status
- **Read-only enforcement** — INSERT, UPDATE, DELETE, DROP, ALTER are blocked by default
- **API token access** — External apps (Python, Power BI, Excel) query via secure tokens
- **On-premise Docker deployment** — Runs entirely inside your own infrastructure

User Roles & Access Control (as per Scope Document Section 5.1)

Role	Who It Is For	What They Can See & Do	What Is Hidden / Blocked
super_admin	Platform owner / IT head	Dashboard, DB Profiles (all), Query Console, Query History (all users), User Management, API Tokens, System settings	Nothing — full access to everything
admin	Department IT admin	Dashboard, DB Profiles (all), Query Console, Query History (all users), User Management	Cannot access API Tokens page; cannot change system-level settings
user (analyst)	Engineer / Analyst staff	Dashboard (read), DB Profiles (own only), Query Console, Query History (own queries only)	Cannot see other users' queries; no User Management; no API Tokens
api_consumer	External application / script	Query execution via API token only — no web UI access	Entire web portal is inaccessible; token scoped to one DB profile

SECURITY POLICY

IMPORTANT — Read-Only Enforcement (Scope §5.6): The following SQL operations are BLOCKED for ALL roles in Phase-1: INSERT • UPDATE • DELETE • DROP • ALTER • TRUNCATE • EXEC Only SELECT / read queries are permitted. This protects production databases.

This guide covers 7 screens:

1	Login Page — authenticate and select your role
2	Dashboard — live platform analytics (admin/super_admin view)
3	DB Profiles — connect and manage database connections
4	Query Console — ask questions in English, get SQL results
5	Query History — full audit log of all queries
6	User Management — create and manage users (admin+ only)
7	API Tokens — generate tokens for external access

1 Login Page

Authenticate into the Vanna AI platform

How to log in:

1	Open browser — navigate to http://localhost
2	The ONGC-branded Sign In page appears
3	Enter your Email Address
4	Enter your Password
5	Click the red Sign In button
6	You are redirected to the Dashboard on success

Demo Login Credentials — Use These to Test:

Login Type	Email	Password	Role	What You Can Access
Super Admin (primary demo)	admin@vanna-platform.local	Admin@ONGC123	super_admin	EVERYTHING — all 7 screens
Your Account	v.karthikeyan.ram@gmail.com	(set during registration)	analyst (user)	Dashboard, DB Profiles (own), Query Console, own History only

HOW TO SEE ROLE-BASED ACCESS IN ACTION

To test role differences: log in as super_admin first (you see Users + API Tokens in sidebar). Then create a new 'user' role account (Step 6), log out, log in as that user — you will notice the Users and API Tokens menu items are hidden for regular users.

What you see on the Login page:

- ONGC logo with 'Energy: Now AND Next' tagline on the left panel
- Feature list: Natural Language to SQL, Role-based access, Multi-DB support, Real-time analytics

- Sign In form — Email, Password, red Sign In button
- Hint box at the bottom showing default credentials for reference
- Link to Create Account for new user self-registration

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Dashboard

Live platform analytics — visible to all logged-in users

ROLE ACCESS

Visible to: super_admin (full), admin (full), user/analyst (read-only stats for own activity) NOT accessible to: api_consumer (no web portal access)

After login you land on the Dashboard. All numbers are fetched live from the database and update automatically as the platform is used.

The 7 Stat Cards (Scope §5.9):

Stat Card	Fresh Install Value	What It Means
Active Users (30d)	2	Users who logged in during the last 30 days
DB Profiles	0	Number of database connections configured
Queries Today	0	Total queries run in the last 24 hours
Active Tokens	0	API tokens currently active
Success Rate	100%	Percentage of queries completed without error
Failed Queries	0	Queries that returned an error or were blocked
Avg Response	0 ms	Average SQL query execution time (last 30 days)

Charts below the stat cards:

- **Query Volume (Last 7 Days)** — Bar chart with daily counts. Red bars = queries ran; flat line = no queries
- **Most Queried Databases** — Horizontal breakdown of query volume by database. Populates after first queries

INFO

Zero values on a fresh install are expected and correct. Once you add a Database Profile (Step 3) and run queries (Step 4), all cards populate with real data.

3 Database Profiles

Connect your enterprise databases (Scope §5.2)

ROLE ACCESS

Visible to: super_admin (all profiles), admin (all profiles), user/analyst (own profiles only) NOT accessible to: api_consumer (web portal not available — they use token-bound profiles)

Each database you want to query must be registered as a Connection Profile. Credentials are stored encrypted. The platform tests the connection before saving.

How to add a new database connection:

1	Click DB Profiles in the left sidebar
2	Click + Add Connection (top right)
3	Enter a Profile Name — e.g. 'ONGC Production PostgreSQL'
4	Select Database Type from the dropdown
5	Enter Host / IP address — e.g. 192.168.1.10
6	Enter Port number (auto-filled based on DB type)
7	Enter Database Name / SID
8	Enter the database Username and Password
9	Toggle Read-Only Mode ON (required for Phase-1 security compliance)
10	Click Save — platform tests the connection and confirms success

Supported Database Types (Scope §3):

#	Database	Default Port	Phase-1 Status
1	PostgreSQL	5432	Full support — Recommended

2	MySQL / MariaDB	3306	Full support
3	Microsoft SQL Server	1433	Full support
4	Oracle Database	1521	Full support (SID or Service Name)
5	MongoDB	27017	Experimental / Limited in Phase-1

Each Profile stores (Scope §5.2):

- Profile name, database type, host/IP, port, database name
- Username and encrypted password
- SSL mode settings (disable / require / verify-ca / verify-full)
- Allowed schemas and allowed tables (restrict what AI can query)
- Read-only mode flag — ON by default for security
- Connection status and last test result

4 Query Console

Ask in English — Vanna AI converts to SQL and runs it (Scope §5.4)

ROLE ACCESS

Visible to: super_admin, admin, user/analyst — ALL logged-in users can run queries NOT accessible to: api_consumer (they query via REST API, not the web console)

The core feature of the platform. Type a question in plain English — Vanna AI generates the SQL, executes it against your database, and returns results as a table.

How to run a query:

1	Click Query Console in the left sidebar
2	Select your database from the dropdown at the top
3	Type your question in the text box — plain English, no SQL needed
4	Click the red Run Query button
5	Vanna AI generates the SQL — shown below your question for transparency
6	The SQL executes on your database — results appear as a table
7	Click Export CSV to download results as a spreadsheet
8	Click Copy SQL to copy the generated SQL to clipboard

Example English questions you can ask:

English Question	What Vanna AI Generates
Show total production by well for the last 30 days	SELECT + GROUP BY with date filter on well_production
List all active wells with production above 1000 barrels	SELECT with WHERE clause and numeric comparison

Count drilling rigs by region	COUNT with GROUP BY region
Show me the top 10 employees by salary	SELECT with ORDER BY salary DESC LIMIT 10
What is the total revenue this month	SUM aggregation with current-month date filter
Show total gatepasses created this month	SELECT COUNT(*) FROM gatepasses WHERE created_at >= ...

Example generated SQL:

GENERATED SQL EXAMPLE

```
SELECT well_name, SUM(production_bbls) AS total_production FROM well_production
WHERE prod_date >= CURRENT_DATE - INTERVAL '30 days' GROUP BY well_name ORDER BY
total_production DESC;
```

READ-ONLY ENFORCEMENT

SECURITY: Destructive queries (INSERT, UPDATE, DELETE, DROP, ALTER, TRUNCATE, EXEC) are automatically blocked for ALL users. Only SELECT queries are permitted in Phase-1.

5 Query History

Complete audit log of all platform queries (Scope §5.8)

ROLE ACCESS — WHAT EACH ROLE SEES

super_admin / admin: see ALL users' queries across the platform user / analyst: see ONLY their own queries — other users' history is hidden api_consumer: no web access — their API queries still appear in admin's view

Every query run on the platform is logged. This page gives complete visibility for auditing, compliance, and debugging. The data is stored permanently.

Information logged for every query (Scope §5.8):

Column	Description
Natural Language Question	The original English question the user typed
Generated SQL	The exact SQL that was produced by Vanna AI
Database Profile	Which database connection was queried
Status	success (green) / failed (red) / blocked (orange)
Execution Time	How long the SQL took to run (milliseconds)
User	Email of the user who ran the query
IP Address	IP address the request came from
Timestamp	Exact date and time of the query
Row Count	How many rows were returned

How to use Query History:

1	Click Query History in the left sidebar (clock icon)
2	Most recent queries appear at the top
3	Click any row to expand and see the full generated SQL

4	Filter by date, user, database, or status to find specific queries
5	Admins can see all users — analysts only see their own queries

6 User Management

Create & manage user accounts and roles (Scope §5.1)

ROLE ACCESS — ADMIN ONLY

Visible to: super_admin (full control), admin (can manage users) HIDDEN from: user/analyst and api_consumer — they cannot see this page at all The 'Users' sidebar link with ADMIN badge is hidden for non-admin roles

User Roles Defined in Scope Document (Section 5.1):

Role	Scope Definition	Sidebar Menus Visible	Hidden From Them
super_admin	Full platform control	All 7 menus: Dashboard, DB Profiles, Query Console, History, Users, API Tokens, Settings	Nothing
admin	Manage users & profiles	Dashboard, DB Profiles, Query Console, Query History (all users), Users	API Tokens page, system settings
user (analyst)	Create / query own profiles	Dashboard, DB Profiles (own only), Query Console, Query History (own only)	User Management, API Tokens, other users' data
api_consumer	Token-based access only	No web portal — API access only	Entire web UI — login page rejects this role

How to create a new user account:

1	Click Users in the left sidebar (only visible to admin / super_admin)
2	Click + Add User (top right)
3	Enter Full Name
4	Enter Email Address — this becomes their login username
5	Set a temporary Password (user should change on first login)

6	Select Role: super_admin / admin / user / api_consumer
7	Click Save — user can log in immediately

How to test role-based access yourself:

1	Log in as super_admin (admin@vanna-platform.local / Admin@ONGC123)
2	Go to Users → Create a new user with role = user, e.g. demo@ongc.local / Demo@ONGC123
3	Log out from the top-right menu
4	Log back in as demo@ongc.local / Demo@ONGC123
5	Notice: Users and API Tokens menu items are GONE from the sidebar
6	Go to Query History — you see only your own queries, not the admin's
7	Go to DB Profiles — you can only see profiles YOU created, not all profiles

Users currently registered in the system:

Full Name	Email	Role	Status
Super Administrator	admin@vanna-platform.local	super_admin	Active
Karthikeyan V	v.karthikeyan.ram@gmail.com	user (analyst)	Active

7 API Tokens

Token-based access for external apps (Scope §5.3 & §6)

ROLE ACCESS — SUPER ADMIN ONLY

Visible to: super_admin only (full control of all tokens) HIDDEN from: admin, user/analyst, api_consumer — they cannot see or generate tokens This protects the token management console from misuse

API Tokens allow external applications — Python scripts, Power BI, Excel, Tableau — to query any connected database without using the web UI. Each token is bound to one specific database profile (Scope §5.3).

Token features (Scope §5.3):

- **Token creation** — generate a unique secure token bound to a DB profile
- **Token revocation** — instantly disable a token if compromised
- **Token expiry** — set 30 days / 90 days / 1 year / Never
- **Token regeneration** — rotate a token while keeping the same profile binding
- **Usage tracking** — see how many requests each token has made
- **Rate limiting** — each token has a per-minute request limit
- **Profile binding** — each token is locked to exactly one database profile

How to generate an API token:

1	Log in as super_admin — API Tokens menu is only visible to this role
2	Click API Tokens in the left sidebar (key icon)
3	Click + Generate Token
4	Give it a descriptive name — e.g. 'HR System MySQL Token'
5	Select the Database Profile this token will be bound to
6	Set an expiry date
7	Click Generate — the token is shown ONCE ONLY — copy it immediately!
8	Store the token securely — it cannot be retrieved after closing the dialog

External application usage (Scope §6.1):

API REQUEST / RESPONSE FORMAT

```
POST http://localhost/api/v1/query/ask Authorization: Bearer YOUR_TOKEN_HERE Request body: { "question": "Show total production by well this month" } Response: { "generated_sql": "SELECT well_name, SUM(...)", "summary": "Total production by well this month", "data": [...] }
```

Digital Assistant examples from Scope Document (§4.4):

Application	Database Type	Token Bound To
HR Assistant	MySQL	MySQL profile — hr_database
Finance Assistant	PostgreSQL	PostgreSQL profile — finance_db
Asset Assistant	SQL Server	MSSQL profile — asset_management

Next Steps — Making the Platform Live

Complete these steps in order to move from fresh install to fully operational:

1	**Add a Database Profile** — DB Profiles → + Add Connection → enter your PostgreSQL / Oracle details → Save
2	**Run your first query** — Query Console → select database → type 'show all tables' or any question → Run Query → verify results
3	**Create team accounts** — User Management → + Add User → create accounts for each staff member with the correct role
4	**Change admin password** — Profile settings → change from Admin@ONGC123 to a strong secure password immediately
5	**Test role isolation** — Create a user-role account, log in as them, confirm they cannot see User Management or API Tokens
6	**Generate API Tokens** — For each external system, create a token bound to its database profile in API Tokens
7	**Verify Dashboard** — After a few queries, confirm stat cards and charts show real data

System Requirements:

- Docker Desktop installed and running (Windows / Mac / Linux)
- Minimum 4 GB RAM allocated to Docker (7 containers run simultaneously)
- Port 80 available on the host machine for web access
- Network connectivity from Docker host to your target databases

Technology Stack (Scope §7):

Component	Technology Used
Frontend	React 18 + Vite + Tailwind CSS
Backend API	FastAPI (Python) — JWT / OAuth2 authentication
AI Engine	Vanna AI — natural language to SQL

Database	PostgreSQL — platform metadata storage
Cache / Queue	Redis — async job processing
Reverse Proxy	Nginx — routes web + API traffic
Containerization	Docker Compose — 7 containers
ORM	SQLAlchemy 2.0 (async)